RExercise#7_Porras

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Part A — Creating Data Frames

Question #1:

```
# TODO: Before combining these into a data frame, confirm all six vectors
# have the same length using length() on each one (or wrap them in c()
# and check for yourself that every value is 20)

# 1. Create the vectors from Figure 1
respondents <- 1:20
sex <- c(2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 1, 2)
fathers_occupation <- c(1, 3, 3, 3, 1, 2, 3, 1, 1, 1, 3, 2, 1, 3, 3, 1, 3, 1, 2, 1)
persons_at_home <- c(5, 7, 3, 8, 5, 9, 6, 7, 8, 4, 7, 5, 4, 7, 8, 8, 3, 11, 7, 6)
siblings_at_school <- c(6, 4, 4, 1, 2, 1, 5, 3, 1, 2, 3, 2, 5, 5, 2, 1, 2, 5, 3, 2)
types_of_houses <- c(1, 2, 3, 1, 1, 3, 3, 1, 2, 3, 2, 3, 2, 2, 3, 3, 3, 3, 3, 2)

# 2. Confirm that each vector has a length of 20
length(respondents)
```

```
## [1] 20
```

```
length(sex)
```

```
## [1] 20
```

```
length(fathers_occupation)
```

```
## [1] 20
```

```
length(persons_at_home)
```

```
## [1] 20
```

```
length(siblings_at_school)
```

```
## [1] 20
```

```
length(types_of_houses)
```

```
## [1] 20
```

```
# 3. Combine vectors into a data frame
survey_data <- data.frame(
  respondents = respondents,
  sex = sex,
  fathers_occupation = fathers_occupation,
  persons_at_home = persons_at_home,
```

```
  siblings_at_school = siblings_at_school,
  types_of_houses = types_of_houses
)

# Display the data frame
survey_data
```

```
##    respondents sex fathers_occupation persons_at_home siblings_at_school
## 1            1   2                  1               5                  6
## 2            2   2                  3               7                  4
## 3            3   1                  3               3                  4
## 4            4   2                  3               8                  1
## 5            5   2                  1               5                  2
## 6            6   2                  2               9                  1
## 7            7   2                  3               6                  5
## 8            8   2                  1               7                  3
## 9            9   2                  1               8                  1
## 10          10   2                  1               4                  2
## 11          11   1                  3               7                  3
## 12          12   2                  2               5                  2
## 13          13   2                  1               4                  5
## 14          14   2                  3               7                  5
## 15          15   2                  3               8                  2
## 16          16   2                  1               8                  1
## 17          17   2                  3               3                  2
## 18          18   2                  1              11                  5
## 19          19   1                  2               7                  3
## 20          20   2                  1               6                  2
##    types_of_houses
## 1                1
## 2                2
## 3                3
## 4                1
## 5                1
## 6                3
## 7                3
## 8                1
## 9                2
## 10               3
## 11               2
## 12               3
## 13               2
## 14               2
## 15               3
## 16               3
## 17               3
## 18               3
## 19               3
## 20               2
```

```
# TODO: Combine all six vectors into a data frame called survey using
# data.frame(), naming the columns: Respondent, Sex, FathersOccupation,
# PersonsAtHome, SiblingsAtSchool, TypesOfHouses
# Print the resulting data frame
```

```r
survey <- data.frame(
  Respondent = respondents,
  Sex = sex,
  FathersOccupation = fathers_occupation,
  PersonsAtHome = persons_at_home,
  SiblingsAtSchool = siblings_at_school,
  TypesOfHouses = types_of_houses
)
survey
```

```
##    Respondent Sex FathersOccupation PersonsAtHome SiblingsAtSchool
## 1           1   2                 1             5                6
## 2           2   2                 3             7                4
## 3           3   1                 3             3                4
## 4           4   2                 3             8                1
## 5           5   2                 1             5                2
## 6           6   2                 2             9                1
## 7           7   2                 3             6                5
## 8           8   2                 1             7                3
## 9           9   2                 1             8                1
## 10         10   2                 1             4                2
## 11         11   1                 3             7                3
## 12         12   2                 2             5                2
## 13         13   2                 1             4                5
## 14         14   2                 3             7                5
## 15         15   2                 3             8                2
## 16         16   2                 1             8                1
## 17         17   2                 3             3                2
## 18         18   2                 1            11                5
## 19         19   1                 2             7                3
## 20         20   2                 1             6                2
##    TypesOfHouses
## 1              1
## 2              2
## 3              3
## 4              1
## 5              1
## 6              3
## 7              3
## 8              1
## 9              2
## 10             3
## 11             2
## 12             3
## 13             2
## 14             2
## 15             3
## 16             3
## 17             3
## 18             3
## 19             3
## 20             2
```

```r
# a. Describe the data. Get the structure or the summary of the data.

str(survey)
```

```
## 'data.frame':    20 obs. of  6 variables:
##  $ Respondent       : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Sex              : num  2 2 1 2 2 2 2 2 2 2 ...
##  $ FathersOccupation: num  1 3 3 3 1 2 3 1 1 1 ...
##  $ PersonsAtHome    : num  5 7 3 8 5 9 6 7 8 4 ...
##  $ SiblingsAtSchool : num  6 4 4 1 2 1 5 3 1 2 ...
##  $ TypesOfHouses    : num  1 2 3 1 1 3 3 1 2 3 ...
```

```r
summary(survey)
```

```
##    Respondent          Sex       FathersOccupation PersonsAtHome
##  Min.   : 1.00   Min.   :1.00   Min.   :1.00      Min.   : 3.0
##  1st Qu.: 5.75   1st Qu.:2.00   1st Qu.:1.00      1st Qu.: 5.0
##  Median :10.50   Median :2.00   Median :2.00      Median : 7.0
##  Mean   :10.50   Mean   :1.85   Mean   :1.95      Mean   : 6.4
##  3rd Qu.:15.25   3rd Qu.:2.00   3rd Qu.:3.00      3rd Qu.: 8.0
##  Max.   :20.00   Max.   :2.00   Max.   :3.00      Max.   :11.0
##  SiblingsAtSchool TypesOfHouses
##  Min.   :1.00     Min.   :1.0
##  1st Qu.:2.00     1st Qu.:2.0
##  Median :2.50     Median :2.5
##  Mean   :2.95     Mean   :2.3
##  3rd Qu.:4.25     3rd Qu.:3.0
##  Max.   :6.00     Max.   :3.0
```

```r
# b. Check for the dimensions. How many rows and columns?

dim(survey)
```

```
## [1] 20  6
```

```r
nrow(survey)
```

```
## [1] 20
```

```r
ncol(survey)
```

```
## [1] 6
```

```r
# c. Create factors for Sex, Father's Occupation and Types of Houses

survey$Sex <- factor(survey$Sex,
                     levels = c(1, 2),
                     labels = c("Male", "Female"))

survey$FathersOccupation <- factor(survey$FathersOccupation,
                                   levels = c(1, 2, 3),
                                   labels = c("Farmer", "Driver", "Others"))

survey$TypesOfHouses <- factor(survey$TypesOfHouses,
                               levels = c(1, 2, 3),
                               labels = c("Wood", "Semi-Concrete", "Concrete"))
```

```r
str(survey)
```

```
## 'data.frame':    20 obs. of  6 variables:
##  $ Respondent       : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Sex              : Factor w/ 2 levels "Male","Female": 2 2 1 2 2 2 2 2 2 2 ...
##  $ FathersOccupation: Factor w/ 3 levels "Farmer","Driver",..: 1 3 3 3 1 2 3 1 1 1 ...
##  $ PersonsAtHome    : num  5 7 3 8 5 9 6 7 8 4 ...
##  $ SiblingsAtSchool : num  6 4 4 1 2 1 5 3 1 2 ...
##  $ TypesOfHouses    : Factor w/ 3 levels "Wood","Semi-Concrete",..: 1 2 3 1 1 3 3 1 2 3 ...
```

```r
# d. Get the mean for all columns.

sapply(survey, function(x) if(is.numeric(x)) mean(x) else NA)
```

```
##        Respondent               Sex FathersOccupation     PersonsAtHome
##             10.50                NA                NA              6.40
##  SiblingsAtSchool     TypesOfHouses
##              2.95                NA
```

```r
# e. Extract the 1st five rows and then all the columns using subsetting functions. What is the output?

survey[1:5, ]
```

```
##   Respondent    Sex FathersOccupation PersonsAtHome SiblingsAtSchool
## 1          1 Female            Farmer             5                6
## 2          2 Female            Others             7                4
## 3          3   Male            Others             3                4
## 4          4 Female            Others             8                1
## 5          5 Female            Farmer             5                2
##   TypesOfHouses
## 1          Wood
## 2 Semi-Concrete
## 3      Concrete
## 4          Wood
## 5          Wood
```

```r
# f. Select the variable types of houses then store the vector that results as types_houses.

types_houses <- survey$TypesOfHouses
types_houses
```

```
##  [1] Wood          Semi-Concrete Concrete      Wood          Wood
##  [6] Concrete      Concrete      Wood          Semi-Concrete Concrete
## [11] Semi-Concrete Concrete      Semi-Concrete Semi-Concrete Concrete
## [16] Concrete      Concrete      Concrete      Concrete      Semi-Concrete
## Levels: Wood Semi-Concrete Concrete
```

```r
# g. Select only all Males respondent that their father occupation was farmer.

survey[survey$Sex == "Male" & survey$FathersOccupation == "Farmer", ]
```

```
## [1] Respondent        Sex               FathersOccupation PersonsAtHome
## [5] SiblingsAtSchool  TypesOfHouses
## <0 rows> (or 0-length row.names)
```

```r
# h. Select only all females respondent that have greater than or equal to 5 number of siblings attendi
```

```r
survey[survey$Sex == "Female" & survey$SiblingsAtSchool >= 5, ]
```

```
##    Respondent    Sex FathersOccupation PersonsAtHome SiblingsAtSchool
## 1           1 Female            Farmer             5                6
## 7           7 Female            Others             6                5
## 13         13 Female            Farmer             4                5
## 14         14 Female            Others             7                5
## 18         18 Female            Farmer            11                5
##    TypesOfHouses
## 1           Wood
## 7       Concrete
## 13 Semi-Concrete
## 14 Semi-Concrete
## 18      Concrete
```

Question #2:

```r
# Before running data.frame(), you were asked to check that all six vectors had the same length. Why is

# Explanation:
# In survey data, each row corresponds to a single respondent's complete set of answers across all ques
# If vectors have unequal lengths, the row alignment breaks-causing R to either throw an error ("argume
```

Question #3:

```r
# All five survey variables (Sex through TypesOfHouses) are stored as plain numbers (1, 2, 3, ...), but
# Looking at the legend, which of these columns are categories being represented by numbers (nominal da
# Explain why treating a category like sex as if it were a quantity to average would be misleading.

# Explanation:
# - Categorical / Nominal Variables: Sex (1=Male, 2=Female), FathersOccupation (1=Farmer, 2=Driver, 3=O
# - Count / Numeric Variables: PersonsAtHome and SiblingsAtSchool (actual counts of people).
# - Why averaging Sex is misleading: The numbers 1 and 2 are arbitrary category codes without numeric v
```

Question #4:

```r
# In subset-logical, if you filter on Sex == 1 using the raw numeric code rather than the word "Male".
# Looking at the result, how could someone unfamiliar with the legend misinterpret this output?
# What problem does this illustrate about working with coded data before it has been labeled?

# Explanation:
# - Misinterpretation: Someone without the legend might assume 1 indicates a numerical quantity (e.g.,
# - Problem Illustrated: Raw numeric codes create ambiguity and make data analysis dependent on externa
```

Question #5:

```r
# Out of 20 respondents, how many are Male (Sex == 1)? Compute this using nrow() on your filtered resul
# Does this small sample size change how confident you would be generalizing any pattern you find in th
```

```r
nrow(survey[survey$Sex == "Male", ])
```

```
## [1] 3
```

```r
# or using raw coded data:
sum(survey$Sex == 1)
```

```
## [1] 0
```

```r
# Explanation:
# Only 3 out of 20 respondents are Male.
# Yes, this small sample size severely reduces confidence in generalizing findings to a larger populati
```

Question #6:

```r
# Run summary(). Every column shows numeric summary statistics (min, mean, max, etc.) - including Sex a
# Explain why R doesn't know, by itself, that these columns are categorical, and what this tells you ab

summary(survey)
```

```
##    Respondent        Sex     FathersOccupation PersonsAtHome  SiblingsAtSchool
##  Min.   : 1.00   Male  : 3   Farmer:9          Min.   : 3.0   Min.   :1.00
##  1st Qu.: 5.75   Female:17   Driver:3          1st Qu.: 5.0   1st Qu.:2.00
##  Median :10.50               Others:8          Median : 7.0   Median :2.50
##  Mean   :10.50                                 Mean   : 6.4   Mean   :2.95
##  3rd Qu.:15.25                                 3rd Qu.: 8.0   3rd Qu.:4.25
##  Max.   :20.00                                 Max.   :11.0   Max.   :6.00
##        TypesOfHouses
##  Wood          : 4
##  Semi-Concrete: 6
##  Concrete      :10
##
##
##
```

```r
# Explanation:
# R evaluates variables based strictly on their technical data type, not their context. Because Sex and
# This illustrates the difference between data storage and semantic meaning: software stores values pur
```

Question #7:

```r
# Compare the mean of PersonsAtHome to the mean of Sex in your summary output. Which of these two means

# Explanation:
# - PersonsAtHome mean (~6.55): Meaningful. This column records actual quantitative counts of people, s
# - Sex mean (~1.85): Not meaningful. Sex is a categorical variable represented by arbitrary code numbe
```